

GT-B159 Safety Orientation

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This document provides an overview of safety protocols for Gates-Thomas B159 (colloquially known as the "Arm Lab").

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1 General Rules

1. Keep your common sense about you
2. Be aware of all robot's workspaces
3. Keep the e-stop within arm's reach at all times
4. Keep food and drink away from the electronics
5. Watch for loose wires on the floor - tape these down when possible
6. Work in the buddy system
7. Tie up loose hair and loose clothing when near robots
8. Never run a robot without a trained person present and aware of your plans

2 In the Event of a Trapped Individual

On the Franka:

1. Attempt to alleviate the E-stop or P-stop and move it in Manual mode.
2. Insert the Red Key into the joint at the base of the Franka, then move the arm as required.

On the UR10e:

1. Attempt to alleviate the E-stop or P-stop that has been triggered. Then move the robot away in Manual mode.

2. Physically backdrive the robot. If needed, yell for assistance or seek assistance from other individuals nearby.
3. Unplug the robot and backdrive it. This will reset the absolute encodings - use in SERIOUS emergencies.

3 Theatrics

Folks, welcome to the final day of your long awaited B159 training programme. It has been a long journey to get here, you should be very proud of yourselves. Make sure to frame this accomplishment on your wall once you have successfully passed.

First things first, a few simple knowledge based questions:

1. Where should the e-stop be?
2. How close can I get to the robot?
3. Johnny is working late at night and brings in a delicious Thanksgiving meal from Red Door. Should he:
 - (a) Toss his food in the trash before entering the lab.
 - (b) Eat his food at the tables away from all electronics.
 - (c) Bring his dinner to the workbench and let the arms feed it to him?
4. What is the most important thing to keep with you in the lab at all times?

Oh boy, we've arrived at the live-action part of your examination. Don't worry, this is only a 50 minute reenactment of your favourite moments from safety training... AGHH... It's got me! I've been pinned by the Franka! What can you do to help me? Oh no! The E-stop is continuing to be triggered! Save me! Save me!

Wow. Glad we made it through... AGHHH... The UR is taking revenge. I've been taken. What can you do? Oh no! Manual isn't working! I'm choking and you can't back drive it alone! How are you going to help? Hurry!

Thanks team. Didn't think I'd be getting such an assist today. You've definitely earned your safety badges. Welcome to the ARM lab.